

Project Manager Maneuver Ammunition Systems

Product Lead Special Ammunition and Weapon Systems

GENERAL SPECIFICATION FOR
PdL SAWS MORTAR WEAPON SYSTEMS
10 February 2026

1. SCOPE

This specification describes the performance requirements and identifies the verification procedures for the Mortar Weapon systems. This specification supports the acquisition of a smooth bore mortar weapon system consisting of a Barrel, Sighting Device, Base plate, Bipod and Basic Issue Items (BII). This specification includes the minimum essential Engineering and Packaging Requirements and the necessary Quality Assurance Provisions to determine that these requirements have been met.

2. APPLICABLE DOCUMENTS AND DEFINITIONS

Order of precedence, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Requirements specified in the contract or delivery order precede the text of this document. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. All documents listed in this specification should use the latest revision available at the time of award.

2.1 Documents

INTERNATIONAL TEST OPERATING PROCEDURES DOCUMENTS

ITOP 4-2-805 Projectile Velocity and Time of Flight Measurements, 20 September 2005.

ITOP 3-2-810 Weapon Chamber Pressure Measurements, 18 December 1985.

ITOP 3-2-803 Visual Inspection of Cannon Bores, 1 October 1992.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION DOCUMENTS

ISO 3452-4 Non-Destructive Testing - Penetrate Testing - Part 4: Equipment - First Edition; Together with ISO 3452-1, ISO 3452-2 and ISO 3452-3 Cancels and Replaces ISO 3452:1984.

ISO DIS 9934-1 Non-destructive testing - Magnetic particle testing - Part 1: General principles.

INTERNATIONAL STANDARDS FOR PHYTOSANITARY MEASURES DOCUMENTS

ISPM-15 International Standards for Phytosanitary Measures No. 15.

2.2 Definitions.

Basic Issue Items (BII): Standard items the manufacturer includes with every mortar weapon system to assist in setup, cleaning and maintenance.

Cleaning Kit: Kit which contains the necessary items to clean the weapon which shall include items such as bore brush, cleaning rod, cleaning solvents, and cleaning cloths.

Maintenance Kit: Kit which includes necessary tools to perform maintenance such as application specific wrenches, screwdrivers, and pliers.

Maintenance Supplies Kits: Kits that include commonly replaced items and spare parts such as springs, firing pins, washers, gaskets, nuts, and bolts.

Sight Glass: The weapon sight used to aim the weapon.

Mils: The TDP shall list the type of optics included with the weapon.

For this document:

The Warsaw Pact mil is defined as an angle comprising 1/6000th of a circle, equal to 0.06 degrees and shall appear with “WP” after mils. i.e. 200 mils (WP).

The NATO mil is defined as an angle comprising 1/6400th of a circle, equal to 0.05625 degrees and shall appear with “NATO” after mils. i.e. 200 mils (NATO).

Aiming equipment: Equipment used to assist the soldiers in aiming the sight to hit the target. Items may include aiming stakes, plotting board, and firing tables.

3. REQUIREMENTS

3.1 General requirements: Mortar weapon system and its associated parts shall conform to the applicable drawings provided by the manufacturer. The mortar weapon systems shall be of new production, within three (3) years of the delivery date, free of corrosion, gouges, dents, cracks, and in good working order. Basic Issue Items (BII) shall be provided.

3.2 Reliability: Each mortar weapon system shall perform the firing, pressure sustaining, projectile launching and accuracy functions. The manufacturer shall use ammunition of their choice which is capable of demonstrating that the mortar weapon system meets all the requirements of this specification. Additionally, the mortar weapon system must be serviceable under conditions including temperature extremes of at least +50°C to -30°C without interference, erratic movement or malfunctions. There shall be no impairment in the functioning of mortar weapon assemblies, component assemblies, or component parts, during and/or after firing. After field maintenance is performed, the mortar weapon system must continue to operate in weather/climatic conditions including snow, rain, mud, salt spray, dust/sandstorm, and solar radiation.

3.3 Maintainability: The manufacturer must ensure the availability of spare parts and BII for the mortar weapon system for a minimum of five (5) years after final delivery. The mortar weapon system must not require special tools other than what is provided by the manufacturers BII.

3.4 Packaging: Packaging and packing shall be in accordance with Section 4.5 of this specification or as otherwise specified in the contract or delivery order.

3.5 Marking: Marking shall be in accordance with Section 4.3.1 of this specification or as otherwise specified in the contract or purchase order.

3.6 Durability: The mortar weapon system shall possess the required durability to withstand the rigors of mounted/dismounted patrol and firing all propellant zones from a variety of ground surfaces; sand, crusher run, loam, and hard packed soil.

3.7 Interchangeability: All barrels, assemblies, sub-assemblies, and parts shall be interchangeable with all other mortar weapon systems regardless of lot and/or subplot.

3.8 Compatibility: The mortar weapon system must be capable of providing the maximum range when compared to the lot history folders of the ammunition chosen for use during the LAT. Meteorological and Topographical Elevation corrections can be applied to the results if necessary.

4. VERIFICATION

4.1 Technical verification. The mortar weapon system shall be subject to all requirements of applicable technical data in addition to verification in accordance with TABLE I of this specification. Noncompliance with any requirements shall be cause to withhold acceptance of the lot or batch in which the noncompliance was found.

TABLE I. Requirements/Conformance Verification [Lot Acceptance Testing (LAT)]

Section 3	Requirements	Section 4 Verification Procedures	Sample Size
3.1	General requirements	4.1	All systems
3.2	Reliability	4.2	All systems
3.3	Maintainability	N/A	N/A
3.4	Packaging	4.5	All systems
3.5	Marking	4.3.1	All systems
3.6	Durability	N/A	N/A
3.7	Interchangeability	4.3.7	All systems
3.8	Compatibility	N/A	N/A

4.2 Live Fire Testing:

4.2.1 Seating the Baseplate.

The Baseplate shall be firmly seated to the manufacturer's satisfaction prior to beginning the proof testing. It is recommended that at least one cartridge be fired at top charge during the seating of the baseplate.

4.2.2 Proof Testing.

All weapons shall be proof tested as a complete system utilizing the associated components accompanying that weapons serial number. To ensure compatibility of each mortar weapon system with its associated ammunition the following Proof Pressure must be met. The values are based on using a copper crusher test method.

Mortar Weapon System	Minimum Proof Pressure kgf/cm ²
60mm Commando	500
60mm	630
82mm	745
120mm	1140

It is recommended that cartridges with inert fill be used during this test, however the decision shall be left to the manufacturer.

Elevation/Traverse: The weapon shall be set to 45 degrees [750 mils (WP), 800 mils (NATO)] elevation. The traverse shall be centered and the system re-leveled.

The cartridge shall be loaded with additional propellant so that its chamber pressure is at least the Proof Pressure. If the barrel fails to reach system Proof Pressure in five (5) attempts, the barrel shall be reinspected for damage, and the Contractor shall notify the U.S. Government (USG) by contract letter to determine the disposition of the barrel.

Should a cartridge achieve a pressure of 5% over the Proof Pressure, the barrel shall be marked in a way to ensure it can be differentiated from the other barrels being tested. The barrel, however, may continue to be ballistically tested and inspected as per this specification. The Contractor shall notify the USG by contract letter, to determine the disposition of the barrel. The barrel shall not be marked with a proof mark until the disposition by the USG to do so is received.

Once a cartridge meets or exceeds the Proof Pressure, the Proof test is complete.

4.2.2.1 Proof Test Report

At the conclusion of the Proof Test a report shall be provided to PdL SAWS. The report shall contain at a minimum the following information:

- a. Part name
- b. Drawing number and revision
- c. Test date
- d. Serial number of weapon(s) tested

- e. Required values
- f. Actual values
- g. Attribute requirements
- h. Attribute requirements verification - Pass/Fail status. If “Fail”, description of failure
- i. Copies of associated test reports, for instance, Ultrasound Test reports
- j. Lot number
- k. Test technician’s/engineer’s name

4.2.3 Traverse Left, Traverse Right, and Elevation Testing.

Test quantity shall be the three (3) weapons or 2% of order quantity, whichever is greater not, but not to exceed five (5) weapons.

The cartridges for this test shall be fired at highest charge that allows for their maximum range. Two (2) cartridges shall be fired for each test, four (4) total.

High Elevation/Left Traverse Test: As the weapon is traversed and elevated, special care shall be taken to note any evidence of stiff, binding, or erratic operation. The weapon shall be set to an elevation of 75 degrees [1250 mils (WP), 1333 mils (NATO)]. The traverse shall be moved to the far left of its adjustable range and the system re-leveled. Two (2) cartridges shall be fired.

Low Elevation/Right Traverse Test: As the weapon is traversed and elevated, special care shall be taken to note any evidence of stiff, binding, or erratic operation. The weapon shall be set to an elevation of 60 degrees [1000 mils (WP), 1067 mils (NATO)]. The traverse shall be moved to the far right of its adjustable range and the system re-leveled. Two (2) cartridges shall be fired.

The traverse and elevation shall be checked and the system re-leveled before each shot.

4.2.4 Weapon Sight and Interface verification (Weapon Accuracy).

Test quantity shall be the three (3) weapons or 2% of order quantity, whichever is greater not, but not to exceed five (5) weapons.

Spotting cartridges shall be used prior to the beginning of the test, however once the test below has begun no additional spotting cartridges are allowed. Every cartridge discharged thereafter shall count toward the pass/fail criteria. With the weapon seated to the satisfaction of the manufacturer, the following tests shall be conducted. All testing, fifteen (15) cartridges total, shall be High Explosive (HE) cartridges set to point detonation mode and shot at maximum charge/range. Between shots within a subtest, only minor leveling/bubble adjustments to the weapon are allowed. Use of sandbags on the base plate and/or bipods is allowed.

Subtest 1: The weapon shall be set to an elevation of 45 degrees [750 mils (WP), 800 mils (NATO)] and the azimuth set to 4.6125 degrees [77 mils (WP), 82mils (NATO)] to the left of center range. Five (5) HE test cartridges shall be discharged with observers noting their respective range and deflection.

Subtest 2: The weapon shall be set to an elevation of 45 degrees [750 mils (WP), 800 mils (NATO)] and the azimuth set to 4.6125 degrees [77 mils (WP), 82 mils (NATO)] to the right of center range. Five (5) HE test cartridges shall be discharged with observers noting their respective range and deflection.

Subtest 3: The weapon shall be set to an elevation of 65.025 degrees [1084 mils (WP), 1156 mils (NATO)] and the azimuth set to 0 mils, center range. Five (5) HE test cartridges shall be discharged with observers noting their respective range and deflection.

Evaluation of Range; The Average Corrected Range (corrected for elevation and meteorological conditions) shall be calculated for Subtest 2 and Subtest 3, from the test data. This average shall fall within the upper and lower limits of the equations below. The respective cartridges' firing table, at the applicable elevation and shall be within:

Ballistic Range Upper Limit (BRUL) = Firing Table Range + [Firing Table Range / 200]

Ballistic Range Lower Limit (BRLL) = Firing Table Range - [Firing Table Range / 200]

$$BRUL \geq \text{Average Corrected Range} \geq BRLL$$

Evaluation of Deflection; The Average Corrected Deflection (corrected for elevation and meteorological conditions) shall be calculated for Subtest 1 and Subtest 2, from the test data and shall be within:

Allowable Corrected Deflection Upper Limit (ACDUL) = Average Corrected Range X sin (4.6125 degrees) + [Average Corrected Range / 400]

Allowable Corrected Deflection Lower Limit (ACDLL) = Average Corrected Range X sin (4.6125 degrees) - [Average Corrected Range / 400]

$$ACDUL \geq \text{Average Corrected Deflection} \geq ACDLL$$

FIRING MATRIX

Section	Subtest	Rounds	Elevation	Cartridge
4.2.1	Seating	1-5	Manufacturers discretion	Any
4.2.2	Proof	1-5	750 mils (WP), 800 mils (NATO)	Inert or HE
4.2.3	Traverse Left and High Elevation	2	1250 mils (WP), 1333 mils (NATO)	HE
4.2.3	Traverse Right and Low Elevation	2	1000 mils (WP), 1067 mils (NATO)	HE
4.2.4	Weapon Sighting (Accuracy)	15	Various	HE

4.3 Post Live Fire inspections.

All weapons used during Live Fire Testing 4.2.3 and 4.2.4 shall be inspected. All weapons used during Proof testing 4.2.2 shall be Magnetic Particle inspected (4.3.2.5). Results shall be provided in the test report.

4.3.1 Marking.

Mortar barrels, assemblies and sub-assemblies shall be clearly and legibly marked to prevent mismatch between different mortar calibers and model types. Such markings shall be in accordance with the requirements and drawings as provided by the manufacturer. Each mortar barrel shall receive a permanent pressure proofing mark/stamp provided it passes all of the Live Fire testing and Magnetic Particle inspection within this document. No mortar barrel shall be shipped or accepted without a permanent pressure proof mark or stamp.

4.3.2 Inspections; Mortar barrel.

4.3.2.1 Firing Pin Protrusion. The protrusion of the firing pin from the bottom of the barrel shall be in accordance with the manufacture's requirements.

4.3.2.2 Interior Dimensions, Leakage and Distortion. Visually inspect the barrel and breech cap for distortion, bulging or cracks. The barrels inner diameter shall be inspected at a minimum of three (3) approximately evenly spaced points over the length of the barrel. At each of these lengths the diameter will be measured twice at 90 degree intervals for a total of six (6) diameter measurements. These measurements shall ensure interior tolerances, as provided by the manufacturer, are met. Additionally, the barrel shall be leak tested to ensure air tightness.

4.3.2.3 Barrel, Interior surface. The barrel bore surface shall be free of material and processing defects such as cracks, seams, tears, flakes, gouges, scratches, corrosion, and pits. The following conditions of the barrel bore surface are acceptable:

- a) A scratch having a depth no greater than 0.0004 inch regardless of its width.
- b) A pit with a depth no greater than 0.001 inch and having a diameter or any other major dimension no greater than 0.030 inch, provided that the total number of pits present throughout the bore does not exceed twenty five (25) and the total number of pits present in any square inch of surface area does not exceed five (5).

Any defects in excess of those listed above, or not covered, must be reported to PdL SAWS for engineering disposition.

4.3.2.4 Barrel Length. The interior barrel length with applicable tolerance shall be per the dimensions provided by the manufacturer or +/- 0.5 cm whichever is smaller.

4.3.2.5 Magnetic Particle Inspection of the Barrel. This inspection shall be conducted after all live fire testing has been completed, as per ISO 3452-1, 2, 3, 4 and ISO DIS 9934-1. The entire exterior surface (360 degrees for the full length of the of the barrel including the end cap) shall be inspected twice by magnetic particle inspection with magnetic fields:

- 1) Parallel to the axis of the barrel.
- 2) Perpendicular to the axis of the barrel.

The magnetic particles shall contain a dye which fluoresces when exposed to Ultraviolet light. Inspection for cracks shall be done in a dark room with the aid of an Ultraviolet light, after being exposed to the magnetic particle solution and while the barrel is being exposed to the magnetic fields.

4.3.3 Inspections; Bipod.

4.3.3.1 Bipod General Inspection. Ensure all applicable parts are present and are unbent. The chain between the bipod legs (if applicable) shall not be broken or show evidence of permanent deformation. Bipod shall fold and unfold without excessive effort.

4.3.3.2 Bipod Cross leveling, Traverse and Elevation testing. The weapon is Cross Leveled, Traversed, and Elevated through its full range of motion(s). During these cross leveling, traverse, and elevation movements, special care is taken to note any evidence of stiff, binding or erratic operation. Additionally, there shall be less than 1/8 of a turn (backlash) of the cross leveling, elevation or traversing hand wheel(s) before movement occurs.

4.3.3.3 Bipod Clamping interface and shock absorber. The bipod shall provide a firm clamping force to the barrel. The barrel shall not slide freely when hung under its own weight. Inspect the bipod/barrel clamp interface for gouges. Ensure there are no bulges on the shock absorber or leaking hydraulic fluids (if applicable). The shock absorber shall be manually cycled to ensure no damage has occurred.

4.3.4 Inspections; Baseplate.

These inspections shall be conducted after all live fire testing has been completed.

4.3.4.1 Baseplate General Inspection. Ensure all applicable parts are present and there is no evidence of permanent deformation and/or cracks. There shall be no binding between the latched ball/socket connection when the barrel is rotated. The base plate shall be retained on the barrel when hung under its own weight via its latched ball/socket connection.

4.3.4.2 Dye Penetrate Inspection of the Baseplate. Each baseplate shall be subjected to a complete liquid dye penetrate (Zyglo) inspection for cracks and trammel point locations will be marked and measured for each baseplate, as per ISO 3452-4 and ISO DIS 9934-1.

4.3.5 Inspections; Weapon sight / Attachment interface. This test verifies there is no significant play or movement between the weapon sight and its weapon mounting interface. The test shall be conducted in accordance with the manufacturer's requirements. An example of how to conduct a weapon sight interface test is included in Appendix A.

4.3.6 Visual Inspections; Weapon and Basic Issue Items (BII). Visual defects shall be identified within the TDP and include a defined method for acceptance/rejection of the mortar weapon and all applicable BII items. Visual inspections listed in ANNEX A shall be the minimum requirement.

4.3.7 Interchangeability. All weapons used during Live Fire Testing 4.2.3 and 4.2.4 shall be broken down into their bipod, base plate, and barrel assemblies. The barrels from each respective system shall be moved one system to the right. The baseplates from each respective system shall be moved one system to the left. The bipods shall remain in their respective locations. All systems shall be capable of being reassembled. If it is not possible to reassemble the systems, the Contractor shall notify the USG by contract letter, to determine the disposition of the systems.

4.4 Documentation. At the completion of mortar verification, the test authority will document all pertinent data regarding the hardware under test/inspection in a report. The report will include the test authorities' recommendation of acceptance or non-acceptance. A copy of the Live Fire Test report, Visual, and Dimensional Inspection report shall be provided for each mortar system (or its major components if purchased separately).

All raw data as well as corrected data (if applicable), collected during live fire testing covered in Section 4.2 shall be provided in the report. The X and Y coordinates of the raw data for each shot, corrected data for each shot, and corrected averages shall be provided using the gun position as the reference point i.e. (0,0).

These reports shall be included with the LAT report.

4.5 Packaging. The weapons shall be delivered with applicable packaging, lubricants, and protectants to ensure the weapon system does not deteriorate during transportation and/or sheltered long term storage for a minimum of ten (10) years at -30°C to +50°C. The systems must be palletized and in accordance with ISPM-15 standards. Steel strapping shall be used to constrain the cargo/weapon system. Dunnage shall be used to ensure the strapping does not damage the palletized hardware.

5. NOTES.

This section contains information of a general nature or explanatory nature that may be helpful but is not mandatory.

5.1 Acquisition requirements.

Acquisition documentation (base award and/or Request for Proposal (RFP)) must specify the following:

- a. Title, number, and date of the specification.
- b. List of drawings, specifications, and publications pertinent to the mortar system technical details and usage; showing revisions and dates of revisions.
- c. Shipping instructions for each mortar, or its major subassemblies (cannons, bipods, and baseplates) if purchased separately.
- d. Serialization requirements should be as specified in corresponding drawings.

5.2 Accuracy of commercial inspection equipment.

When commercial and modified commercial inspection equipment is used, it should be capable of repetitive measurements to accuracy of 10 percent of the total tolerance of the characteristic that is measured.

5.3 Accuracy of proof pressure measuring devices.

Each crusher gage that is used for measuring proof firing pressures should have an accuracy that meets the maximum error of measurement as specified in NATO AEP-23.

5.4 Calibration.

All inspection and test equipment should be calibrated in accordance with the requirements of NCSL Z540.3 or ISO 10012. Proof of calibration should be provided upon demand.

5.5 Performance characteristics inspection and verification.

All requirements should be verified with approved equipment, specified tests, examinations, and procedures.

5.6 Materials and geometric configuration.

Complete and accurate substantiating documentation supporting the acceptability of each mortar weapon system and the components thereof should be compiled and maintained by the manufacturer for the life of the individual weapon. Failure to comply with the drawing requirements, unless specifically waived in the contract, should be cause for rejection.

5.7 Nonconformance.

If any assembly, component, or test specimen fails to comply with any of the requirements of this specification, that unit should be removed, and the actions in MIL-STD-1916 for nonconforming product should be enforced. No repair is authorized unless repair work instructions are approved by the USG.

5.8 Alternative inspection provisions.

Alternative conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, may be used by the Contractor when they provide as a minimum the level of quality assurance required by the provisions specified herein. Prior to applying methods or equipment, the Contractor should describe them in a written proposal submitted to the USG for evaluation. When required, the Contractor should demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. All approved alternative provisions should be specifically incorporated into the Contractor's quality program or detailed inspection system, as applicable.

Frederick Willecke
Product Lead Special Ammunition and Weapon Systems
Picatinny Arsenal, NJ

Annex A: Visual Inspection Requirements

APPENDIX A – Sample Weapon Sight / Attachment Interface Test

ANNEX A – Mortar Weapon and BII Visual Inspection Requirements

Critical: 5-0-1	Major: 5-1-2 Re-inspection: 15-3-4 (cumulative)	Minor (5-3-4) Re-inspection: 15-9-10 (cumulative)
Missing, cracked, loose or bent parts	Cases with moving components; loose pack	Excessive grease, oil or dirt
Gross corrosion, pitting, or deformation	Missing, illegible or incorrect marking	Minor surface corrosion
Missing proof mark	Wire security seal missing	Dents, chips, scratches or burrs
Proof mark present on non-proof tested weapon	Incorrect non-critical components	Paint damaged, wet or incomplete
Incorrect critical components consisting of base plate, bipod, mortar tube, sight, or firing pin (i.e. 60mm base plate supplied on 120mm mortar weapon system)		
Note 1: If the number of allowable defects is exceeded a double sample quantity shall be reinspected. The accept/reject criteria is based on the cumulative number of defects for all three samples. Note 2: If a critical is found or the number of allowable majors or minors, after reinspection, is exceeded the lot fails. Note 3: The inspection and reinspection quantity listed is selected during LAT. DCMA will utilize their Zero-Based Acceptance Sample Plans shown in Appendix B during GSI.		

APPENDIX A - Weapon Sight / Attachment Interface verification

Securely latch the sight to its respective weapon sight interface.

Step 1: Insert the sight unit into its respective interface. Anchor the bipod in a vise, sandbags or similar manner to ensure its immobility. With the bipod at center traverse, aim the right side of the vertical sight unit reticle onto the left side of the blue rectangle on the Weapon Sight Verification Target. The distance between the purple and blue rectangles on the Weapon Sight Verification Target (Step 3) is 3 centimeters across and placed 100 meters from the sight.

Step 2: Level the sight unit and lock the bipod traverse and sight unit into place.

Step 3: The user, while looking through the sight unit, will press forward, with one hand, on the cantilevered sight unit along the line of fire. The force should remove all free play from the sight unit in its respective interface. While looking through the sight the right side of the cross hair shall not pass the right edge of the light blue rectangle. At this range, a 1.5 cm deflection equals 0.14 mils (WP), 0.15 mils (NATO) of deflection.

Step 4: With the bipod at center traverse, aim the left side of the sight unit reticle onto the right side of the dark blue rectangle on the Weapon Sight lot Target (Step 6).

Step 5: Level the sight unit and lock the bipod traverse and sight unit into place.

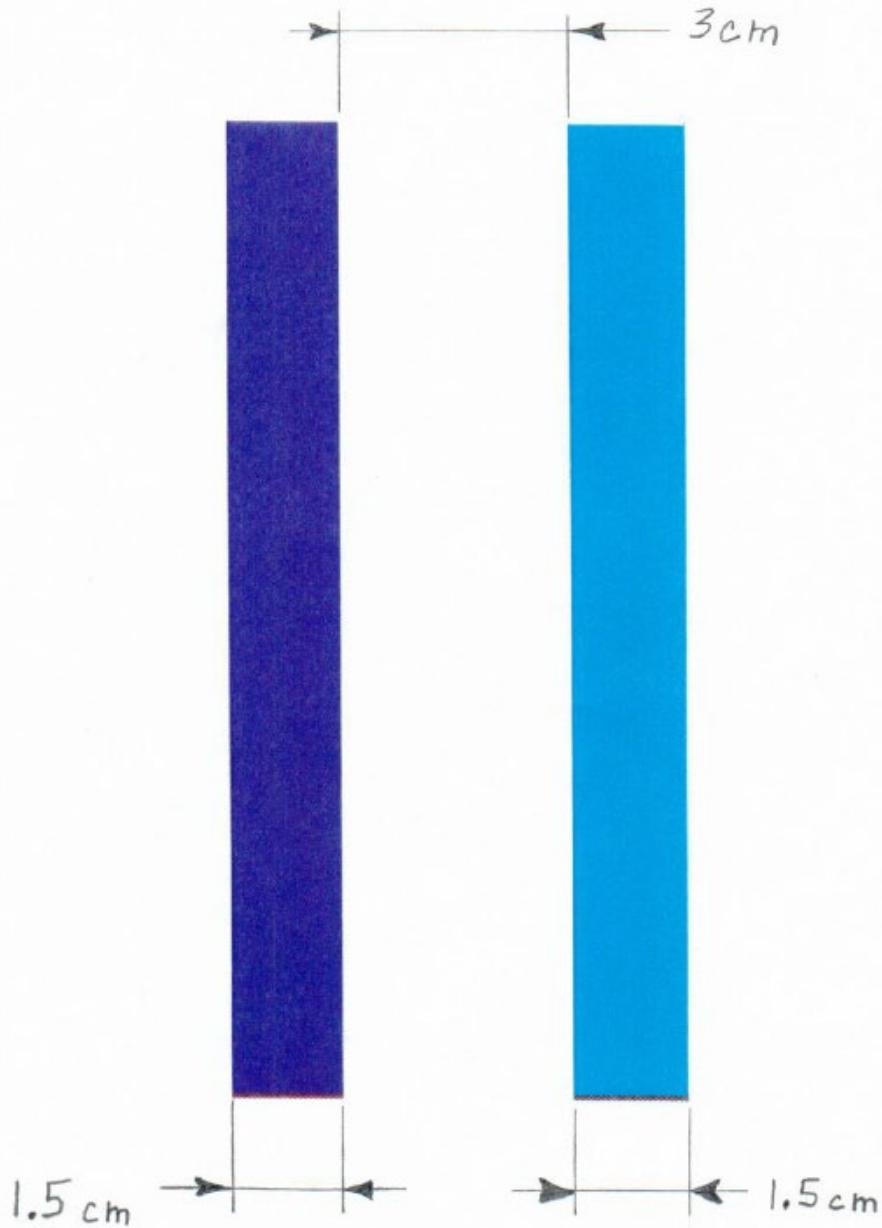
Step 6: The user, while looking through the sight unit, will press backwards, with one hand, on the cantilevered sight unit along the line of fire. The force should remove all free play from the sight unit in its respective interface. While looking through the sight the left side of the cross hair shall not pass the left edge of the dark blue rectangle. At this range, a 1.5 cm deflection equals 0.14 mils (WP), 0.15 mils (NATO) of deflection.

If the “play” in the sight unit interface exceeds 0.14 mils (WP), 0.15 mils (NATO) in either direction, the results must be reported to PdL SAWS for engineering disposition.

APPENDIX A

Weapon Site Verification Target

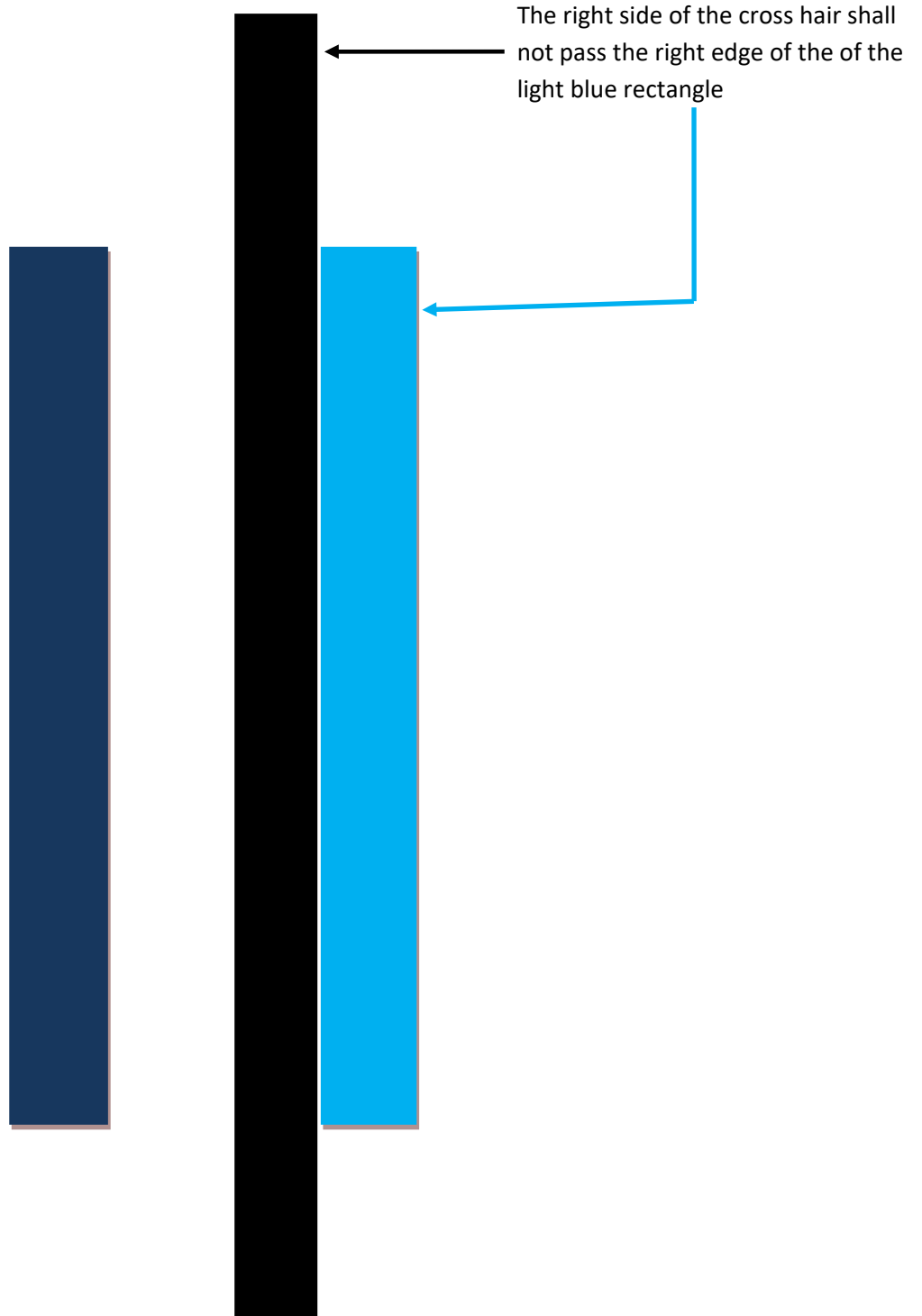
Post this target 100 meters from weapon site being tested



APPENDIX A

Weapon Sight Verification Target

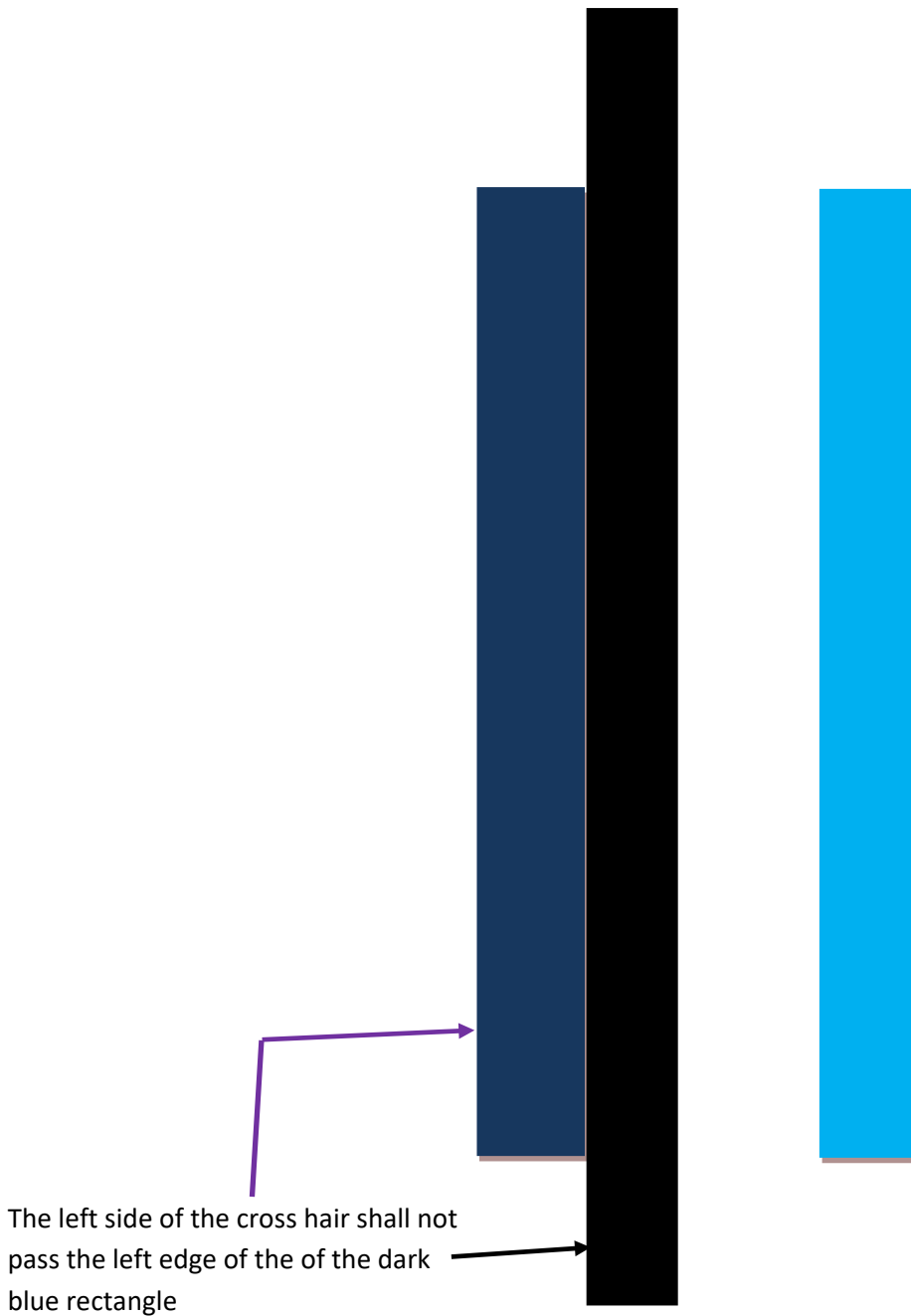
Step 3 (Pass / Fail)



APPENDIX A

Weapon Sight Verification Target

Step 6 (Pass / Fail)



APPENDIX B

	Acceptable Quality Level (AQL)															
LOT SIZE	0.01	0.015	0.025	0.04	0.065	0.10	0.15	0.25	0.40	0.65	1.0	1.5	2.5	4.0	6.5	10.0
1-8	A	A	A	A	A	A	A	A	A	A	A	A	5	3	3	3
9-15	A	A	A	A	A	A	A	A	A	A	13	8	5	3	3	3
16-25	A	A	A	A	A	A	A	A	A	20	13	8	5	3	3	3
26-50	A	A	A	A	A	A	A	A	32	20	13	8	7	7	5	3
51-90	A	A	A	A	A	A	80	50	32	20	13	13	11	8	5	4
91-150	A	A	A	A	A	125	80	50	32	20	19	19	11	9	6	5
151-280	A	A	A	A	200	125	80	50	32	29	29	19	13	10	7	6
281-500	A	A	A	315	200	125	80	50	48	47	29	21	16	11	9	7
501-1200	A	800	500	315	200	125	80	75	73	47	34	27	19	15	11	8
1201-3200	1250	800	500	315	200	125	120	116	73	53	42	35	23	18	13	9
3201-10,000	1250	800	500	315	200	192	189	116	86	68	50	38	29	22	15	9
10,001-35,000	1250	800	500	315	300	294	189	135	108	77	60	46	35	29	15	9
35,001-150,000	1250	800	500	490	476	294	218	170	123	96	74	56	40	29	15	9
150,001-500,000	1250	800	750	715	476	345	270	200	156	119	90	64	40	29	15	9
500,001 & Over	1250	1200	1112	715	556	435	303	244	189	143	102	64	40	29	15	9

Note: An "A" in place of a number indicates that all units in a lot must be inspected